

Introduction:

Research Questions:

- Where are burglary hotspots located in Austin, TX?
- What patterns emerge when comparing burglary rates and proximity to currently available affordable housing?

Project Objectives:

- Using ArcGIS and GIS tools, I want to:
 - Analyse burglary data and identify burglary hotspots
 - Investigate patterns in burglary rates and if it is within 500ft of currently available affordable housing

Background and Importance:

This project aims to analyze burglary hotspots in Austin, Texas, and determine whether they are within 500 feet of currently available affordable housing. By mapping and spatially analyzing these datasets in ArcGIS Pro, I hope to identify potential correlations between crime patterns and affordable housing locations. This analysis may provide insights into safety concerns near affordable housing areas and help inform urban planning decisions.

Data and Methods:

Datasets:

- Crime Data: <https://maps.austintexas.gov/GIS/CrimeViewer/>
 - Format: .shp (Export.zip)
 - Source: Austin Crime Viewer Website
- Affordable Housing:
<https://www.arcgis.com/apps/mapviewer/index.html?layers=dadfcf469e714a508973e02f5ca4781e>
 - Format: ArcGIS Online Map (importable using url)
 - Source: ArcGIS Online Map Viewer

Processing Steps:

To collect the necessary data, I first reviewed the sources I identified during my project proposal phase. I began by examining the Affordable Housing dataset and found that it could be directly imported

into my ArcGIS Pro project using its URL, as it is already hosted on ArcGIS Online. This made the integration straightforward and eliminated the need for manual downloads or conversions.

Next, I retrieved burglary incident data from the Austin Crime Viewer website. Since the tool allows for exporting crime data, I selected a dataset covering one month of burglary incidents and downloaded it as a .zip file containing a shapefile. The exported file structure and naming conventions were pre-determined by the source, so I renamed it to SYA_Final_Project.

GIS Tools and Methods:

To conduct the analysis, I first used the Buffer tool to generate a 500-foot radius around each affordable housing location. This created a defined area where burglary incidents could be examined in relation to affordable housing properties. I set the buffer distance to 500 US Survey feet, and once the buffer zones were established I used the Clip tool to extract only the burglary incidents that occurred within these areas. This step allowed me to isolate relevant crime data and quantify the number of burglaries near affordable housing.

Results and Maps:

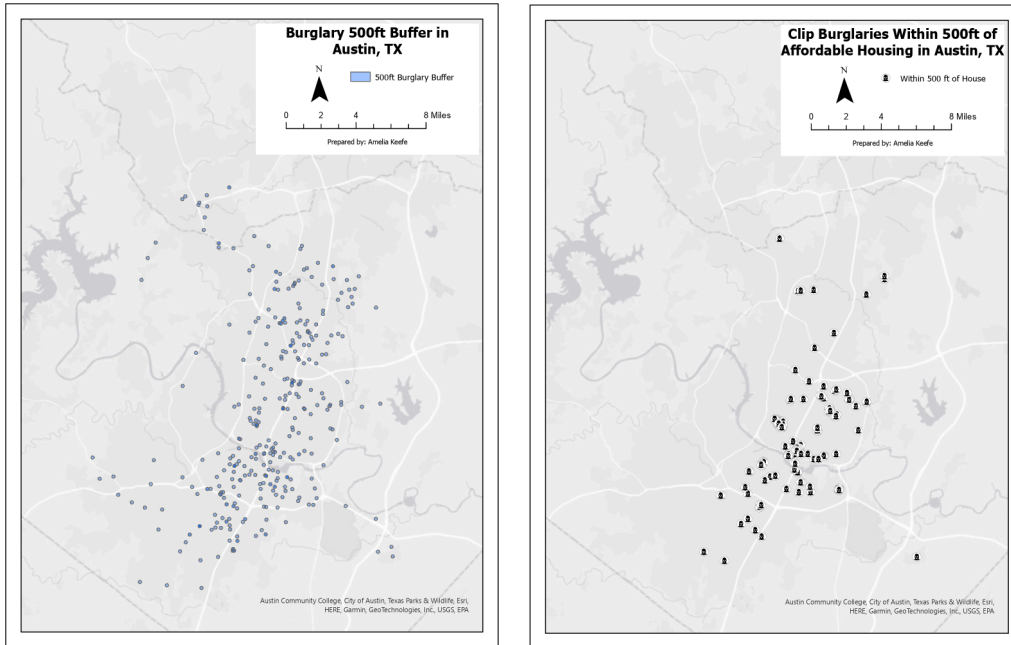
Analysis Results:

The results of my analysis revealed a significant overlap between burglary incidents and affordable housing locations. I found that 251 burglaries occurred within 500 feet of affordable housing properties in Austin during just one month (January 2025). This was an unexpectedly high number, suggesting that certain neighborhoods may be more vulnerable to property crime. While the data does not establish a direct cause-and-effect relationship, it raises important questions about the factors contributing to these patterns, such as socioeconomic conditions, neighborhood design, or law enforcement presence.

This analysis highlights a concerning trend regarding crime rates near affordable housing in Austin. Urban planners, law enforcement, and policymakers could use this information to explore potential solutions, such as increasing security measures, implementing community outreach programs, or improving city planning to reduce crime in these areas.

Visuals:

	Within 500ft of Affordable Housing	Not Within 500ft of Affordable Housing	Total
Burglaries	251	99	350



Conclusion and Discussion:

Summary of Findings:

Overall, this project provided valuable insight into the relationship between crime and affordable housing in Austin. The data indicates a need for increased awareness and action to address the high rate of burglaries in these communities.

Limitations:

The limitations of this project are mainly the time constraint I placed on burglaries. I figured that if I made the time any longer than a month there would be too much data for my computer to handle, causing application crashes and data loss.

Further Research:

Given the results, further research is necessary to determine whether this trend persists over multiple months or if January 2025 was an anomaly. Using additional spatial techniques, such as Hot Spot Analysis, could help identify specific burglary clusters and prioritize areas for intervention.

Bibliography:

Geocortex Viewer for HTML5. (n.d.). <https://maps.austintexas.gov/GIS/CrimeViewer/>

Map Viewer. (n.d.).

<https://www.arcgis.com/apps/mapviewer/index.html?layers=dadfcf469e714a508973e02f5ca4781e>